

## UNIT-V

### PROJECT MANAGEMENT

**Project:** A Project is an organized endeavor to accomplish a specified non-routine or low volume task. Although projects are not repetitive, they take significant amount of time to complete and are large scale or complex enough to be organized and managed as separate undertakings.

**Project management** involves more than just planning, it also requires controlling, monitoring progress and taking corrective action when activities deviate from schedules or costs get out of line. Progress reporting helps managers control by showing cost variances (actual versus budgeted) and time variance (actual versus scheduled) during the projects.

**Importance of Project Management:** Organizations need project management because project management

1. Ensures that customer requirements are met
2. Eliminates duplication of work
3. Reduce the number of tasks that could be overlooked during the project
4. Maximizes the use of resources and
5. Ensures that projects are in control both in terms of project completion time and cost

#### **Problems in Managing a Project:**

1. Managing a project can be a complex and challenging assignment.
2. Since projects are one-of-a-kind endeavors, there may be little in the way of experience, normal working relationships or established procedures to guide participants.
3. A project managers may have to co-ordinate diverse efforts and activities to achieve the project goals.
4. Persons from various disciplines and from various parts of the organization who have never worked together may be assigned to the project for differing spans of time.
5. Sub-contractors who are unfamiliar with the organisation may be brought in to carry out major portions of the project.
6. The project may involve a large number of inter-related activities performed by persons employed by any one of several different sub-contractors.

For the above reasons, it is important that the project managers have an effective means of identifying and communicating the planned activities and the way in which they are to be controlled. Network scheduling methods such as, PERT and CPM have proven to be highly effective and valuable tools during both the planning and execution phases of projects.

#### **NETWORK ANALYSIS:**

Network analysis refers to a number of techniques for the planning and control of complex projects. The basis for network planning is the representation of sequential relationships between activities by means of a network of line. The idea is to link the various activities in such a way that the overall time and cost spend on the project is kept to a minimum.

The various network techniques are

1. Programme Evaluation and Review Technique (PERT)
2. Critical Path Method (CPM)

3. Resource Allocation and Mult project – Scheduling (RAMS)
4. Manpower Allocation Procedure (MAP)
5. Resource Planning and Scheduling Method (RPSM)
6. Least Cost Scheduling (LCS)
7. Project Control System (PCS)
8. Graphical Evaluation Review Technique (GERT)
- 9 Line of Balance Method (LOB)

The two most frequently used networks are PERT and CPM. PERT and CPM are two important planning, scheduling and controlling network techniques. These are essentially schematic models and consists of flow diagrams showing the activities which are required along with their inter relationships to complete a project.

**CPM (Critical Path Method)** was the discovery of M. R. Walker of E. I. Du Pont de Nemours & Co. and J. E. Kelly of Remington Rand, circa 1957. The computation was designed for the UNIVAC-I computer. The first test was made in 1958, when CPM was applied to the construction of a new chemical plant. In March 1959, the method was applied to maintenance shut-down at the Du Pont works in Louisville, Kentucky. Unproductive time was reduced from 125 to 93 hours.

**PERT (Project Evaluation and Review Technique)** was devised in 1958 for the POLARIS missile program by the Program Evaluation Branch of the Special Projects office of the U.S. Navy, helped by the Lockheed Missile Systems division and the Consultant firm of Booz-Allen & Hamilton. The calculations were so arranged so that they could be carried out on the IBM Naval Ordinance Research Computer (NORC) at Dahlgren, Virginia.

The methods are essentially network-oriented techniques using the same principle. PERT and CPM are basically time-oriented methods in the sense that they both lead to determination of a time schedule for the project. The significant difference between two approaches is that the time estimates for the different activities in CPM were assumed to be deterministic while in PERT these are described probabilistically. These techniques are referred as project scheduling techniques.

#### **Application of PERT and CPM:**

- Construction of projects such as building, highways, houses or bridges
- Preparation of bids and proposals for large projects such as multipurpose projects
- Maintenance and planning of oil refineries, ship repairs and other such as large operations
- Development of new weapon systems and new products and services
- Manufacture and assembly of large items such as aero planes or ships repairs and other such as large operations
- Simple projects such as home remodeling housekeeping or painting and so on.

**Basic Steps in PERT / CPM:** Project scheduling by PERT / CPM consists of four main steps

1. Planning
2. Scheduling
3. Allocation of Resources
4. Controlling

## **NETWORKING CONVENTIONS:**

A network uses circles and arrow to represent the planned relationships among the activities required to complete a project. Either of the two conventions can be used to develop the network. They are:

1. Activity on Node Convention (AON)
2. Activity on Arrow Convention (AOA)

### **Rules of Activity on Arrow (AOA) Network Construction:**

The following are some of the rules which have to be followed while constructing activity on arrow AON networking.

1. In a networking diagram arrow represents activities and circles (nodes) represents events. The length of an arrow is of no significance
2. Each activity shall be represented by only one arrow and must start from an event in a circle called node
3. The event numbered one denotes start of the project and this is called initial event. All activities emerging from even numbered one should not be preceded by any other activity or activities, Event carrying the highest number denotes the completion of a project
4. A networking should have only one initial event and one terminal (end) event.
5. A head event is represented by  $i$  and a tail event is represented by  $j$ . The general rule for numbering the events is that the head event should always be numbered larger than the tail event. i.e., events should be numbered such that for each activity  $i < j$
6. An activity must be uniquely identified by its starting event and end event which implies that:
  - a. Event number should not be repeated or duplicated
  - b. Two activities should not be identified by the same set of starting and end events
7. Activities must be represented by their symbols or the corresponding ordered pair of starting and completion events
8. The logical sequence between the activities must follow these rules:
  - a. An event can't occur until all the incoming activities into it have been completed
  - b. An activity cannot start unless all the proceeding activities on which depends have been completed
  - c. A dummy activity does not consume either any resource or time, even though it has to follow the above rules

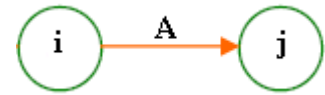
## **NETWORK TERMINOLOGY:**

**1. Event:** An event represents the beginning and or end of some activities. Every activity begins and ends with an event. This is usually represented by a circle in a network which is also called a node or connector.



**2. An Arrow:** It is commonly used to represent an activity indicating the direction of progress in the project. (→)

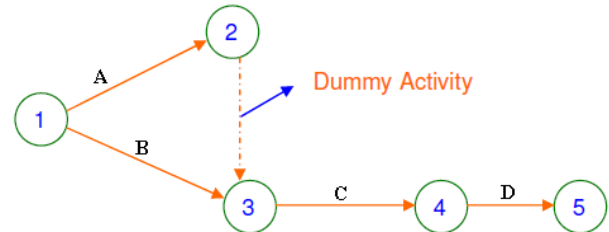
**3. Activity:** Any individual operation which utilizes resources and has an end and a beginning is called activity. An arrow is commonly used to represent an activity with its head indicating the direction of progress in the project. (i, j) Activity with tail event i and head event j.



These are classified into following categories:

**3.1. Predecessor Activity:** Activities that must be completed immediately prior to the start of another activity are called predecessor activities.

**3.2. Successor Activity:** Activities that cannot be started until one or more of other activities are completed but immediately succeed them are called successor activities.



**3.3. Dummy Activity:** An activity that consumes no time and resources, but shows precedence relationship between activities is called as dummy activity. It is represented by dotted arrow.

**In the above example** for activities A&B, activities C&D are succeeding activities and for activity D, C&B activities are preceding activities. For activity C, B is preceding activity and D is succeeding activity

**3.4. Critical Activity:** In a network diagram critical activities are those which if delayed the project will be delayed. Critical activities are shown with thick arrow. (→)

**3.5. Non-critical activity:** Those activities whose delay cannot delay the project completion are called non-critical activities. Such activities have a provision of float or slack

**4. Activity duration:** It is the amount of time an activity consumes for its completion. In CPM it is the best time estimate and in PERT it is the expected or average time to complete an activity.

**5. Network:** Network is a graphical representation of all the project activities showing their interrelationship. It is a logical picture formed after incorporating activities and events. It is also called Arrow diagram.

**6. Critical Path:** The longest path (start to finish) of the network diagram is called critical path. It is the project competition time. This would be the minimum period in which the project can be completed. There may be one or more critical paths in a network. critical path duration is longer than that of any other path in the network.

Any delay in activity(s) on the critical path delays the completion of the project. Hence activities on the critical path require special attention for timely completion. Activities falling on critical path are called critical activities.

**7. Earliest Times:** Times calculated by moving Forward pass i.e., from first to the last event in a network diagram are called earliest times. By the time of forward calculations, in case of alternatives consider the maximum value.

**7.1. Earliest Start Time (EST):** It is the time by which an activity can be started as early as possible. For the first activity or activities in the project **EST=0**.

For all other activities earliest finish of the preceding activity is the earliest start of current activity. In case of multiple preceding activities, EST of current activity = maximum of earliest finish time of preceding activities

**7.2. Earliest Finish Time (EFT):** It is the time by which an activity can be finished or completed as early as possible.

**EFT = Earliest Start Time + Duration of the activity.**

**8. Lates Times:** Times calculated by moving backward pass i.e., from the last event to the first event of the network diagram are called latest times. By the time of backward calculations, in case of alternatives consider the minimum value.

**8.1. Latest Start Time (LST):** It is the time by which an activity can be started as late as possible without delaying the project completion time.

**LST = Latest Finish Time – Duration of the activity**

**8.2. Latest Finish Time (LFT):** It is the time by which an activity can be completed as late as possible without delaying the project completion time. For last activity in the project,

**LFT = LFT.** In case of multiple Earliest Finish Times, Maximum of EFT shall be taken.

For remaining activities **LFT = Latest Start of the Succeeding activity**

**Float or Slack:** Floats or Slacks in the network analysis represent the difference between the maximum time available to finish the activity and the time required to complete it. i.e., reserve time available to complete the activity.

The basic difference between slack and float times is that a slack is used with PERT and float is used with CPM.

**9. Floats are three types:** 1) Total float      2) Free float      3) Independent float

**9.1. Total float:** It is the additional time which a non-critical activity can consume without increasing the project duration. However total float may affect the floats in previous and subsequent activities.

**Total float = LST – EST or LFT – EFT**

**9.2. Free float:** Free float refers to the time by which an activity can expand without affecting succeeding activities.

**Free float = EST of Head Event – EST of Trail Event – Activity duration**

**9.3. Independent float:** This the time by which activity may be delayed or extended without affecting the preceding or succeeding activities in any way.

**Independent float = EST of Head event – LFT of Trail event – Activity duration**

### **PROGRAMME EVALUATION AND REVIEW TECHNIQUE (PERT)**

PERT is a tool to evaluate a given Programme and review the progress made in it from time to time. The main objective in the analysis through PERT is to find out the completion for a particular event within specified date. PERT is probabilistic in nature and takes into account the uncertainties. In PERT three-time values are associated with each activity. They are:

1. **Optimistic time:** It is the shortest possible time in which the activity can be finished. It is the minimum time an activity takes for its completion assuming that everything goes well i.e., there will not be any disturbance such as delay, setbacks and so on. This is denoted by  $t_o$ .

2. **Most likely time:** It is the estimate of the normal time the activity would take. This assumes normal delays. If a graph is plotted in the time of completion and the frequency of completion in that time period, then most likely time will represent the highest frequency of occurrence. This is denoted by  $t_m$ .

3. **Pessimistic time:** This is the maximum possible time for completing an activity if everything goes wrong or in abnormal conditions. This is denoted by **tp**.

PERT do not consider any one of the above three-time estimates, but PERT is based on average time estimate of the above three called as expected time.

**Expected time:** It is the average time an activity will take if it were to be repeated on large number of times and is based on the assumption that the activity time follows Beta distribution. It is denoted by **te** and calculated as  $t_e = (t_o + 4 t_m + t_p) / 6$ .

### PROBLEMS:

1. A small engineering project consists of six activities. The three-time estimates in number of days for each activity are given below.

| Activity | to | tm | tp |
|----------|----|----|----|
| 1-2      | 2  | 5  | 8  |
| 2-3      | 1  | 1  | 1  |
| 3-5      | 0  | 6  | 18 |
| 5-6      | 7  | 7  | 7  |
| 1-4      | 3  | 3  | 3  |
| 4-5      | 2  | 8  | 14 |

**Find out:**

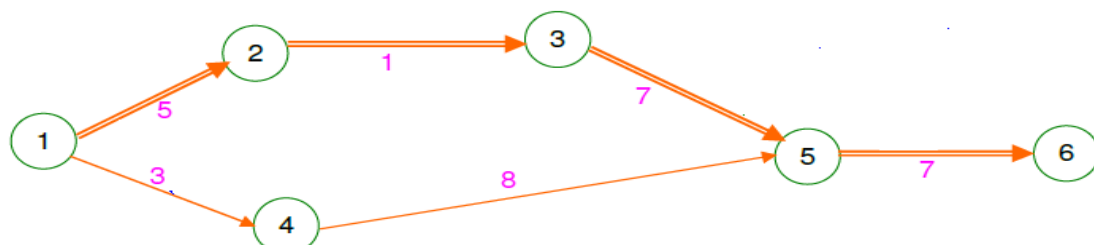
1. Calculate expected time of each activity
2. Draw the network diagram and mark te on each activity
3. Calculate EST&EFT and LST&LFT and mark them on the network diagram
4. Identify the critical path and mark on the network diagram
5. Calculate total slack for each activity
6. Find out the Probability of completing project in 25 days.

**Solution:**

**Calculation of Expected Time, Standard deviation and Variance**

| Activity | t <sub>o</sub> | t <sub>m</sub> | t <sub>p</sub> | $t_e = \frac{t_o + 4t_m + t_p}{6}$ | Variance = $\frac{t_p - t_o}{6}$ | Variance <sup>2</sup> |
|----------|----------------|----------------|----------------|------------------------------------|----------------------------------|-----------------------|
| 1-2      | 2              | 5              | 8              | 5                                  | 1                                | 1                     |
| 2-3      | 1              | 1              | 1              | 1                                  | 0                                | 0                     |
| 3-5      | 0              | 6              | 18             | 7                                  | 3                                | 9                     |
| 5-6      | 7              | 7              | 7              | 7                                  | 0                                | 0                     |
| 1-4      | 3              | 3              | 3              | 3                                  | 0                                | 0                     |
| 4-5      | 2              | 8              | 14             | 8                                  | 2                                | 4                     |

**Construction of Network diagram**



**The different ways to reach the end of the project are**

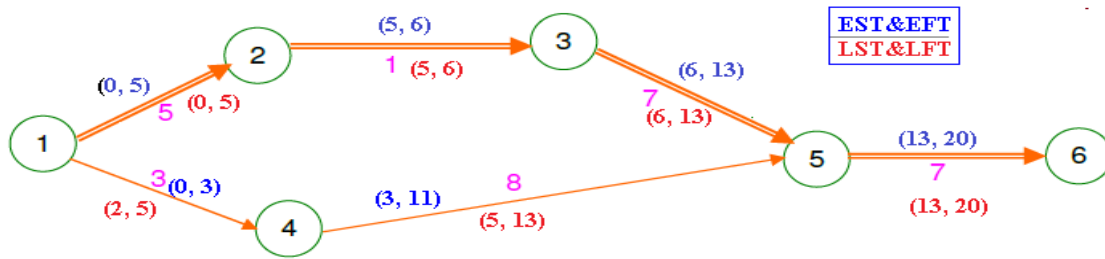
$$1-2-3-5-6 = 5+1+7+7 = 20 \text{ days}$$

$$1-4-5-6 = 3+8+7 = 18 \text{ days}$$

**Critical path is 1-2-3-5-6 = 20 days**

**Project duration is 20 days**

## Determination of Earliest and Latest times and Slack



| Activity | EST | EFT | LST | LFT | Slack |
|----------|-----|-----|-----|-----|-------|
| 1-2      | 0   | 5   | 0   | 5   | 0     |
| 2-3      | 5   | 6   | 5   | 6   | 0     |
| 3-5      | 6   | 13  | 6   | 13  | 0     |
| 5-6      | 13  | 20  | 13  | 20  | 0     |
| 1-4      | 0   | 3   | 2   | 5   | 2     |
| 4-5      | 3   | 11  | 5   | 13  | 2     |

Slack = LST – EST or LFT – EFT

### Probability of completing the project in 25 days:

It is assumed that the probability of completing the project in 20 days = 0.5 or 50%

Probability of completing the project in 25 days is >50%

To find out the probability of completing the project in 25 days, we use normal distribution.

Probability of project completion can be calculated using the following equation

$$\text{Probability (Z)} = \frac{t_s - t_e}{\sigma t_e}$$

Where,  $t_s$  = Duration within which project is to be completed = 25 days

$t_e$  = Duration of the critical path = 20 days

$\sigma t_e$  = Standard deviation of critical path =  $\sqrt{\text{variance of critical path}}$   
 $= \sqrt{1+0+9+0} = \sqrt{10} = 3.16$

$$\text{Probability (Z)} = \frac{t_s - t_e}{\sigma t_e} = \frac{25 - 20}{3.16} = 1.58$$

As per Z table, Z value of 1.58 = 0.9429 = 94.29%

**Probability of completing the project in 25 days is 94.29%**

2. Consider the project consisting of 7 jobs with the following precedence relation and three-time estimates.

| Job | Predecessor | $t_o$ | $t_m$ | $t_p$ |
|-----|-------------|-------|-------|-------|
| A   | -           | 2     | 5     | 8     |
| B   | A           | 6     | 9     | 12    |
| C   | A           | 5     | 14    | 17    |
| D   | B           | 5     | 8     | 11    |
| E   | C, D        | 3     | 6     | 9     |
| F   | -           | 3     | 12    | 21    |
| G   | E, F        | 1     | 4     | 7     |

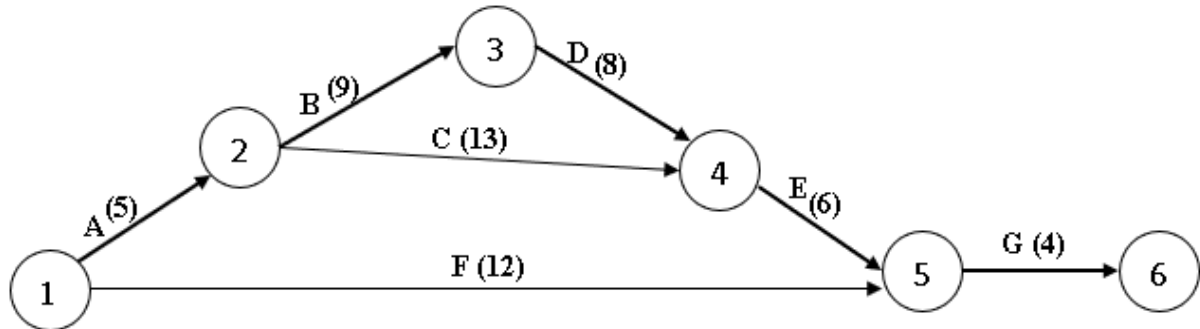
Draw the network, find the critical path, find project duration and find the project duration at 95% probability.

**Solution:**

**Calculation of expected times, standard deviation and variance:**

| Job | $t_o$ | $t_m$ | $t_p$ | $t_e$ | Variance<br>$= t_p - t_o / 6$ | Variance <sup>2</sup> |
|-----|-------|-------|-------|-------|-------------------------------|-----------------------|
| A   | 2     | 5     | 8     | 5     | 1                             | 1                     |
| B   | 6     | 9     | 12    | 9     | 1                             | 1                     |
| C   | 5     | 14    | 17    | 13    | 2                             | 4                     |
| D   | 5     | 8     | 11    | 8     | 1                             | 1                     |
| E   | 3     | 6     | 9     | 6     | 1                             | 1                     |
| F   | 3     | 12    | 21    | 12    | 3                             | 9                     |
| G   | 1     | 4     | 7     | 4     | 1                             | 1                     |

**Construction of Network diagram**



**The different ways to reach the end of the project are**

1-2-3-4-5-6 = 5+9+8+6+4 = 32 days

1-2-4-5-6 = 5+13+6+4 = 28 days

1-5-6 = 12+4 = 16 days

Longest path of the network is critical path.

**critical path is 1-2-3-4-5-6 = 5+9+8+6+4 = 32 days  
Project duration is 32 days**

**Project duration at 95% Probability =?**

$$\text{Probability (Z)} = \frac{t_s - t_e}{\sigma t_e}$$

Where,

$t_s$  = Duration within which project is to be completed =?

$t_e$  = Duration of the critical path = 32 days

$\sigma t_e$  = Standard deviation of critical path =  $\sqrt{\text{variance of critical path}} = \sqrt{5} = 2.236$

Z = Probability factor = 1.65

$$1.65 = \frac{t_s - 32}{2.236}$$

By cross multiplying the above equation, we get  $t_s = 35.6$  days

Project duration at 95% probability is **35.6 days**



3. Consider a project consisting of 7 jobs with the following precedence relationship and 3-time estimates

| Job | Preceding Activities | $t_o$ | $t_m$ | $t_p$ |
|-----|----------------------|-------|-------|-------|
| A   | -                    | 2     | 5     | 8     |
| B   | A                    | 6     | 9     | 12    |
| C   | A                    | 5     | 14    | 17    |
| D   | B                    | 5     | 8     | 11    |
| E   | C, D                 | 3     | 6     | 9     |
| F   | -                    | 3     | 12    | 21    |
| G   | E, F                 | 1     | 4     | 7     |

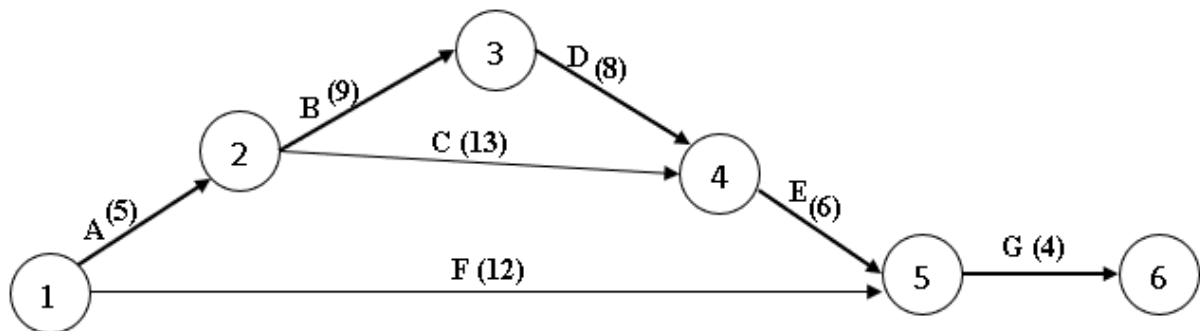
1. Draw the Network
2. Determine expected time and standard deviation of each job.
3. Compute the probability of completing the project 2 days earlier than expected and 4 days later than expected.

**Answer:**

**Calculation of expected time and standard deviation:**

| Job | Preceding Activities | $t_o$ | $t_m$ | $t_p$ | $t_e$ | Variance = $\frac{t_p - t_o}{6}$ | Variance <sup>2</sup> |
|-----|----------------------|-------|-------|-------|-------|----------------------------------|-----------------------|
| A   | -                    | 2     | 5     | 8     | 5     | 1                                | 1                     |
| B   | A                    | 6     | 9     | 12    | 9     | 1                                | 1                     |
| C   | A                    | 5     | 14    | 17    | 13    | 2                                | 4                     |
| D   | B                    | 5     | 8     | 11    | 8     | 1                                | 1                     |
| E   | C, D                 | 3     | 6     | 9     | 6     | 1                                | 1                     |
| F   | -                    | 3     | 12    | 21    | 12    | 3                                | 9                     |
| G   | E, F                 | 1     | 4     | 7     | 4     | 1                                | 1                     |

**1. Network construction:**



Various paths of the network are:

$$A-B-D-E-G = 5+9+8+6+4 = 32 \text{ Days}$$

$$A-C-E-G = 5+13+6+4 = 28 \text{ Days}$$

$$F-G = 12+4 = 16 \text{ Days}$$

**Critical Path is A-B-D-E-G = 5+9+8+6+4 = 32 Days**

**Project duration is 32 Days**

### 3(a) Probability of completing the project 2 days earlier than expected =?

Project expected duration is 32 days.

2 days earlier than expected mean 32days – 2 days = 30 days

**Probability of completing the project in 30 days =?**

$$\text{Probability (Z)} = \frac{t_s - t_e}{\sigma t_e}$$

Where,

$t_s$  = Duration within which project is to be completed = 30 days

$t_e$  = Duration of the critical path = 32 days

$\sigma t_e$  = Standard deviation of critical path =  $\sqrt{\text{variance of critical path}} = \sqrt{5} = 2.236$

$$Z = \frac{30 - 32}{2.236} = \frac{-2}{2.236} = -0.8944$$

As per Z table, Z value of -0.8944 = 0.1867 = 18.67%

**Probability of completing the project in 30 days = 18.67%**

### 3(b) Probability of completing the project 4 days later than expected =?

Project expected duration is 32 days.

4 days later than expected mean 32days + 4 days = 36 days

**Probability of completing the project in 36 days =?**

$$\text{Probability (Z)} = \frac{t_s - t_e}{\sigma t_e}$$

Where,

$t_s$  = Duration within which project is to be completed = 36 days

$t_e$  = Duration of the critical path = 32 days

$\sigma t_e$  = Standard deviation of critical path =  $\sqrt{\text{variance of critical path}} = \sqrt{5} = 2.236$

$$Z = \frac{36 - 32}{2.236} = \frac{4}{2.236} = 1.79$$

As per Z table, Z value of 1.79 = 0.9633 = 96.33%

**Probability of completing the project in 36 days = 96.33%**

## CRITICAL PATH METHOD

Critical path method assumes that the time required to complete an activity can be predicted fairly accurately, and thus, the costs involved can be quantified once the critical path has been identified. Since time is an important factor, CPM involves a trade-off between costs and time. It involves determining an optimum duration for the project, that is, a minimum duration that involves the lowest overall costs.

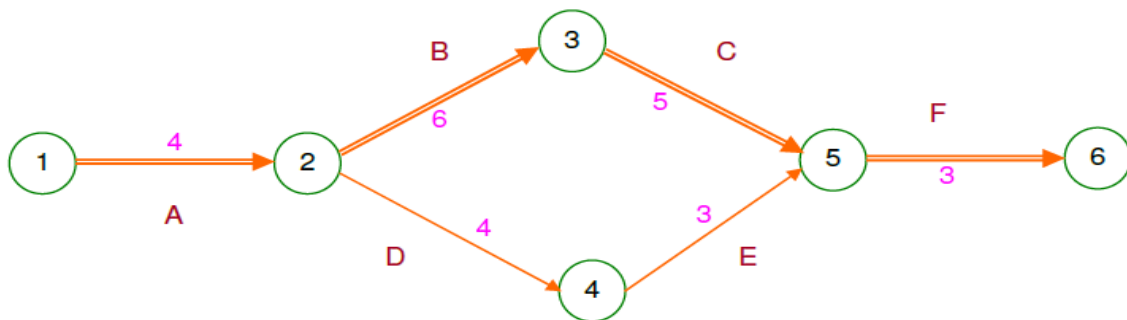
### PROBLEMS

1. A small engineering project consists of 6 activities namely ABCDE & F with the following Duration and precedence relationship. Draw the network diagram and calculate EST, LST, EFT, LFT and floats. Mark the critical path and find total project duration.

| Activity           | A | B | C | D | E | F    |
|--------------------|---|---|---|---|---|------|
| Preceding Activity | - | A | B | A | D | C, E |
| Duration           | 4 | 6 | 5 | 4 | 3 | 3    |

**Solution:**

#### **Construction of Network diagram**



#### **Determination of critical path:**

Various paths of the network are:

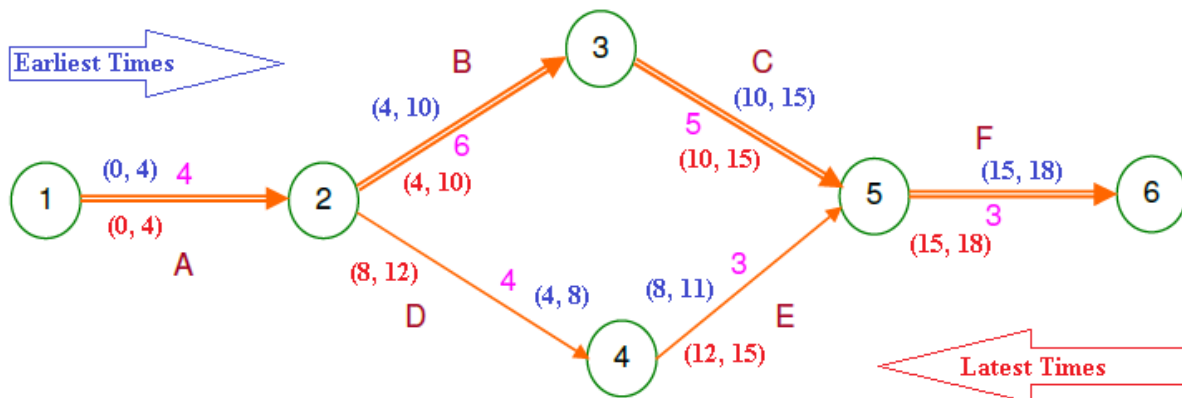
$$1-2-3-5-6 = 4+6+5+3 = 18 \text{ days}$$

$$1-2-4-5-6 = 4+4+3+3 = 14 \text{ days}$$

Critical Path is 1-2-3-5-6 = 4+6+5+3 = 18 days

**Project Duration = 18days**

#### **Determination of EST, EFT, LST & LFT and Calculation of Floats:**



| Activity | Duration | EST | EFT | LST | LFT | Total Float | Free Float | Independent Float |
|----------|----------|-----|-----|-----|-----|-------------|------------|-------------------|
| A        | 4        | 0   | 4   | 0   | 4   | 0           | 0          | 0                 |
| B        | 6        | 4   | 10  | 4   | 10  | 0           | 0          | 0                 |
| C        | 5        | 10  | 15  | 10  | 15  | 0           | 0          | 0                 |
| D        | 4        | 4   | 8   | 8   | 12  | 4           | 0          | 0                 |
| E        | 3        | 8   | 11  | 12  | 15  | 4           | 4          | 0                 |
| F        | 3        | 15  | 18  | 15  | 18  | 0           | 0          | 0                 |

Total float = LST – EST or LFT – EFT

Free float = EST of Head Event – EST of Tail Event – Activity duration

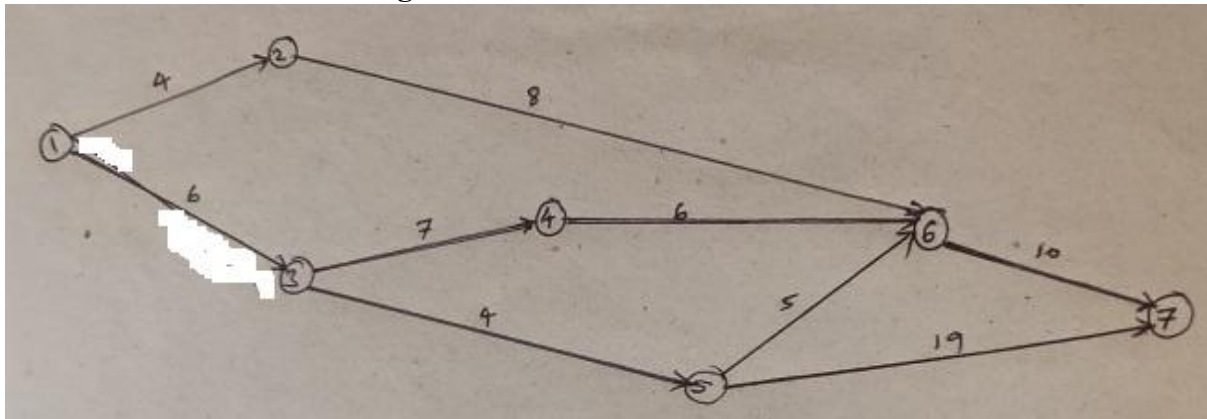
Independent float = EST of Head event – LFT of Tail event – Activity duration

2. A small aircraft project consists of following activities. Draw the network diagram & calculate EST, LST, EFT, LFT and floats. Mark the critical path and find total project duration.

| Events   | 1-2 | 1-3 | 2-6 | 3-4 | 3-5 | 4-6 | 5-6 | 5-7 | 6-7 |
|----------|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Duration | 4   | 6   | 8   | 7   | 4   | 6   | 5   | 19  | 10  |

### Solution:

#### **Construction of Network diagram:**



#### **Identification of critical path:**

The different ways to reach the end of the project (i.e., start to end or left to right) are

$$1-2-6-7 = 4+8+10 = 22 \text{ days}$$

$$1-3-5-6-7 = 6+4+5+10 = 25 \text{ days}$$

$$1-3-4-6-7 = 6+7+6+10 = 29 \text{ days}$$

$$1-3-5-7 = 6+4+19 = 29 \text{ days}$$

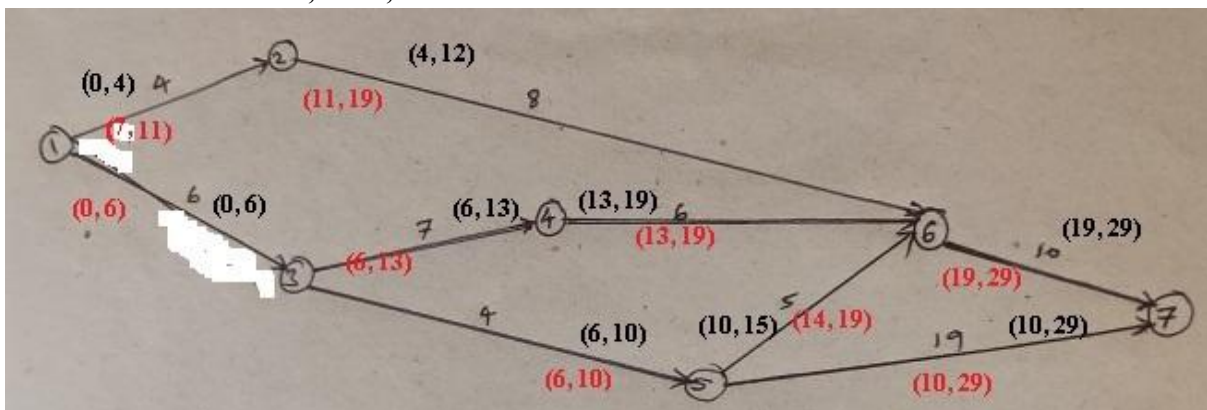
#### **Critical Paths are:**

$$1-3-4-6-7 = 6+7+6+10 = 29 \text{ days}$$

$$1-3-5-7 = 6+4+19 = 29 \text{ days}$$

Project Duration is 29 days

#### **Determination of EST, EFT, LST & LFT and Calculation of Floats:**



| Activity   | Duration | EST | EFT | LST | LFT | Total Float | Free Float | Independent Float |
|------------|----------|-----|-----|-----|-----|-------------|------------|-------------------|
| 1-2        | 4        | 0   | 4   | 7   | 11  | 7           | 0          | 0                 |
| <b>1-3</b> | 6        | 0   | 6   | 0   | 6   | 0           | 0          | 0                 |
| 2-6        | 8        | 4   | 12  | 11  | 19  | 7           | 7          | 0                 |
| <b>3-4</b> | 7        | 6   | 13  | 6   | 13  | 0           | 0          | 0                 |
| <b>3-5</b> | 4        | 6   | 10  | 6   | 10  | 0           | 0          | 0                 |
| <b>4-6</b> | 6        | 13  | 19  | 13  | 19  | 0           | 0          | 0                 |
| 5-6        | 5        | 10  | 15  | 14  | 19  | 4           | 4          | 4                 |
| <b>5-7</b> | 19       | 10  | 29  | 10  | 29  | 0           | 0          | 0                 |
| <b>6-7</b> | 10       | 19  | 29  | 19  | 29  | 0           | 0          | 0                 |

### **PROJECT CRASHING**

**Project Crashing** can be defined as reducing the total project duration by putting extra effort and cost. Crashing is also called as expediting. Project Crashing results in reduced duration and increased cost.

Project cost can be defined as the cost required to complete the project. Project cost can be divided into two. They are

**Direct Cost:** Direct costs are those, which are proportional to the number of activities involved in the project. These costs are variable in nature. Ex. Materials, labour, fuel, etc.,

**Indirect Cost:** Those costs which are fixed or not proportional to the number of activities involved in the project. These costs are determined per day. Ex. Depreciation, supervision.

The sum of Direct cost and Indirect cost of the project refers to Total cost of the project.

#### **OTHER TERMINOLOGY:**

**Normal Time (N<sub>T</sub>):** It is the time required for an activity to complete the project under normal circumstance at normal cost.

**Crash Time (C<sub>T</sub>):** It is the minimum time required to complete an activity of a project by employing additional resources. An activity duration cannot be reduced beyond crash time.

**Normal Cost (N<sub>C</sub>):** Normal cost is the cost that is incurred if the project is allowed to take its normal time considering the efficient utilization of resources.

**Crash Cost (C<sub>C</sub>):** It is the additional cost incurred to reduce an activity duration by employing all possible means like overtime, additional machinery, proper materials etc.,

**Cost Slope:** It is the amount that has to be spent over and above the normal cost for reducing the duration by one unit of time.

$$\text{Cost slope} = \frac{\text{Crash cost} - \text{Normal cost}}{\text{Normal time} - \text{Crash time}} \quad S = \frac{C_C - N_C}{N_T - C_T}$$

#### **Steps in Project Crashing:**

1. Calculate cost slope (crashing cost) for all activities in the project.
2. Activities on the critical path only be selected for crashing.
3. Identify and crash critical activity that has minimum crash cost. In the event of multiple critical paths, an activity from each critical path is to be selected for crashing.
4. After project crashing, the network is to be revised (re-drawn) by adjusting the time of the crashed activity(s).
5. The new critical is to be identified again and apply the step 3.
6. This process is to be continued till no more crashing of the project is possible and identify optimum project duration and cost.

## **SOLVED PROBLEMS:**

1. A project has 75 activities and the related information given below

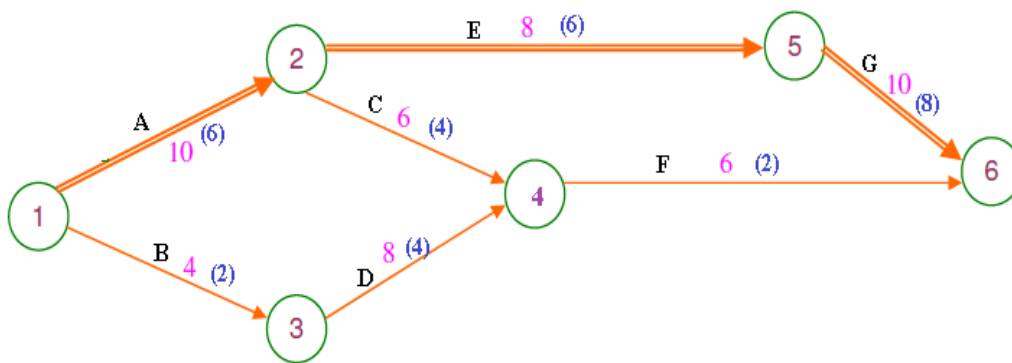
| Activity | Events | Normal Time<br>(days) | Crash Time<br>(days) | Normal<br>Cost | Crash<br>Cost |
|----------|--------|-----------------------|----------------------|----------------|---------------|
| A        | 1-2    | 10                    | 6                    | 400            | 600           |
| B        | 1-3    | 4                     | 2                    | 100            | 140           |
| C        | 2-4    | 6                     | 4                    | 360            | 440           |
| D        | 3-4    | 8                     | 4                    | 600            | 900           |
| E        | 2-5    | 8                     | 6                    | 840            | 1100          |
| F        | 4-6    | 6                     | 2                    | 200            | 300           |
| G        | 5-6    | 10                    | 8                    | 1200           | 1400          |

Indirect cost works out to be 100 per day.

Find out the optimum project cost and optimum project duration

### **Solution:**

#### **Network Construction:**



Number ways to reach the end of the project are:

1-2-5-6 = 10+8+10 = 28 days

1-2-4-6 = 10+6+6 = 22 days

1-3-4-6 = 4+8+6 = 18 days

**Critical path is 1-2-5-6 = 10+8+10 = 28 days**

**Min. duration of the project = 6+6+8 = 20 days** (The project can be crashed up to 20 days)

#### **Calculation of Cost Slope or Crash Cost:**

| Activity | Events     | Normal<br>Time | Crash<br>Time | Normal<br>Cost | Crash<br>Cost | Cost Slope = $\frac{C_c - N_c}{N_T - C_T}$ | Priorities |
|----------|------------|----------------|---------------|----------------|---------------|--------------------------------------------|------------|
| <b>A</b> | <b>1-2</b> | 10             | 6             | 400            | 600           | <b>50</b>                                  | 1          |
| B        | 1-3        | 4              | 2             | 100            | 140           | 20                                         |            |
| C        | 2-4        | 6              | 4             | 360            | 440           | 40                                         |            |
| D        | 3-4        | 8              | 4             | 600            | 900           | 75                                         |            |
| <b>E</b> | <b>2-5</b> | 8              | 6             | 840            | 1100          | <b>130</b>                                 | 3          |
| F        | 4-6        | 6              | 2             | 200            | 300           | 50                                         |            |
| <b>G</b> | <b>5-6</b> | 10             | 8             | 1200           | 1400          | <b>100</b>                                 | 2          |

### **CRASHING:**

Activities on the critical path(s) can only be considered for crashing.

Activity with lowest cost slope is to be selected for crashing.

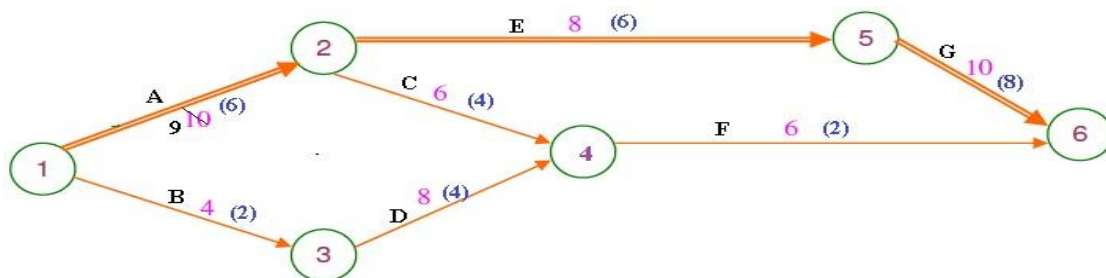
Activities available for crashing are A, E & G.

Activity A has lowest cost slope and it is selected for crashing

### **Crashing Table:**

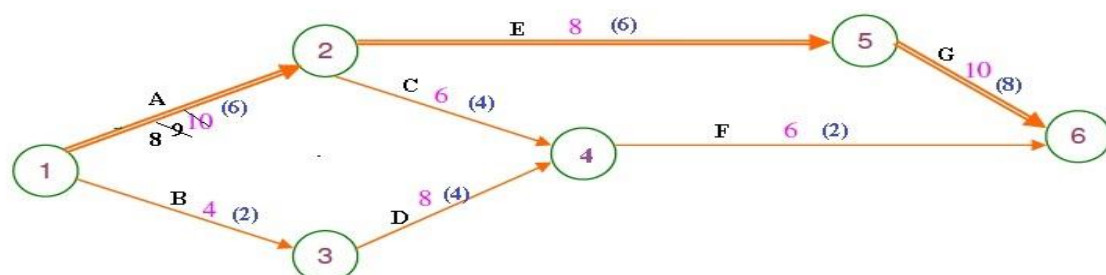
| Crashing No. | Activities available for crashing | Cost slope       | Decision | Project duration | Critical Path |
|--------------|-----------------------------------|------------------|----------|------------------|---------------|
| 1            | A<br>E<br>G                       | 50<br>130<br>100 | A        | 27 days          | 1-2-5-6       |
| 2            | A<br>E<br>G                       | 50<br>130<br>100 | A        | 26 days          | 1-2-5-6       |
| 3            | A<br>E<br>G                       | 50<br>130<br>100 | A        | 25 days          | 1-2-5-6       |
| 4            | A<br>E<br>G                       | 50<br>130<br>100 | A        | 24 days          | 1-2-5-6       |
| 5            | E<br>G                            | 130<br>100       | G        | 23 days          | 1-2-5-6       |
| 6            | E<br>G                            | 130<br>100       | G        | 22 days          | 1-2-5-6       |
| 7            | E                                 | 130              | E        | 21 days          | 1-2-5-6       |
| 8            | E                                 | 130              | E        | 20 days          | 1-2-5-6       |

### **Crashing-1:** Activity-A crashing by 1 day



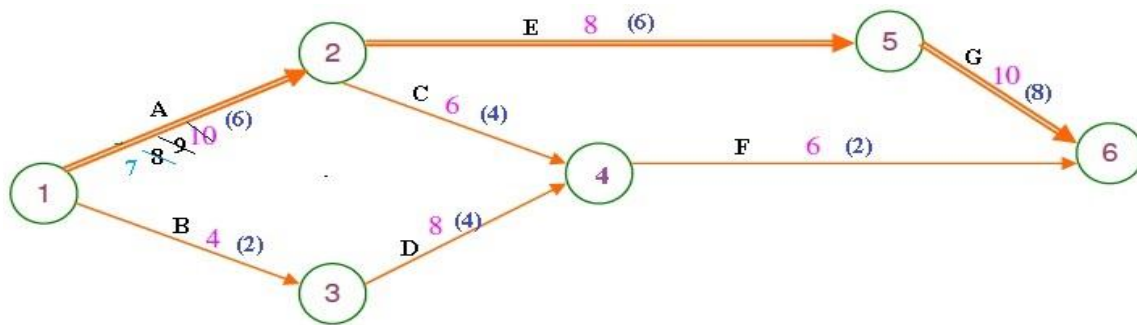
New critical path is 1-2-5-6 and Duration is  $9+8+10 = 27$  days

### **Crashing-2:** Activity-A crashing by 1 another day (Total 2days)



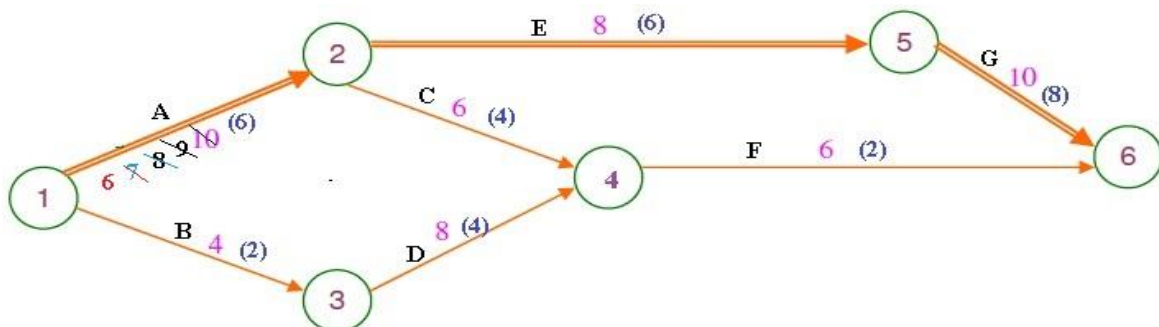
New critical path is 1-2-5-6 and Duration is  $7+8+10 = 26$  days

**Crashing-3:** Activity-A crashing by 1 another day (Total 3days)



New critical path is 1-2-5-6 and Duration is  $7+8+10 = 25$  days

**Crashing-4:** Activity-A crashing by 1 another day (Total 4days)

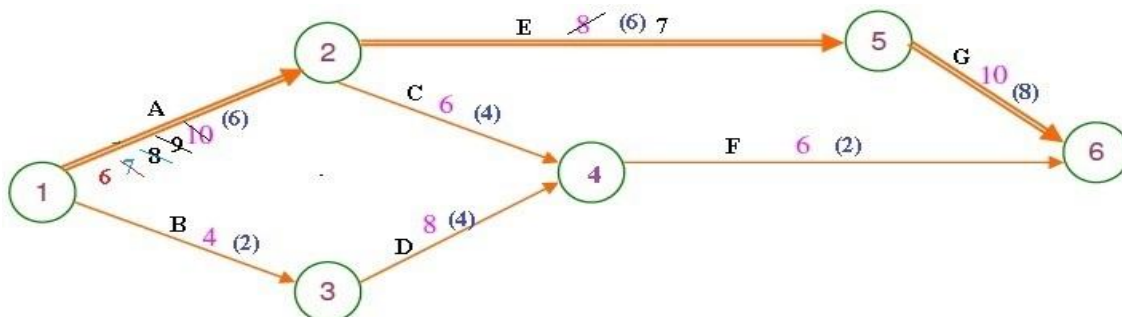


New critical path is 1-2-5-6 and Duration is  $6+8+10 = 24$  days

Further Activity A cannot be crashed as it is reached to a minimum crash duration i.e., 6days

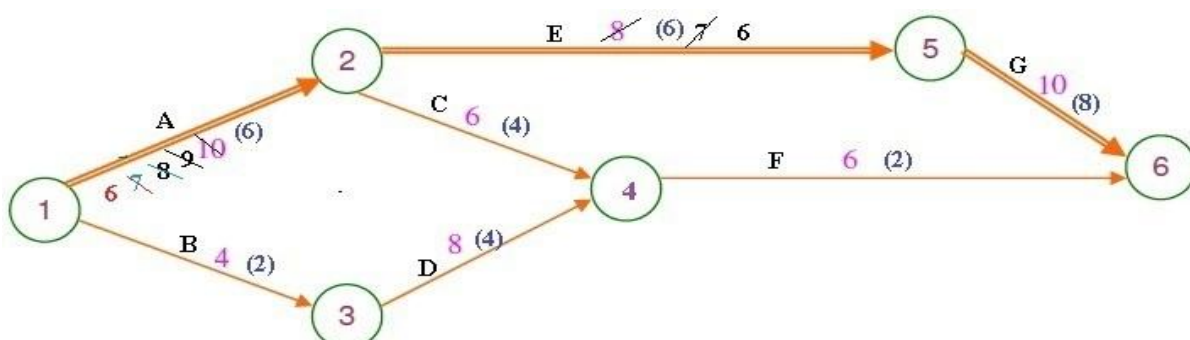
Between activities E & G activity G has lowest cost slope.

**Crashing-5:** Activity-E crashing by 1 day



New critical path is 1-2-5-6 and Duration is  $6+7+10 = 23$  days

**Crashing-6:** Activity-E crashing by 1 another day



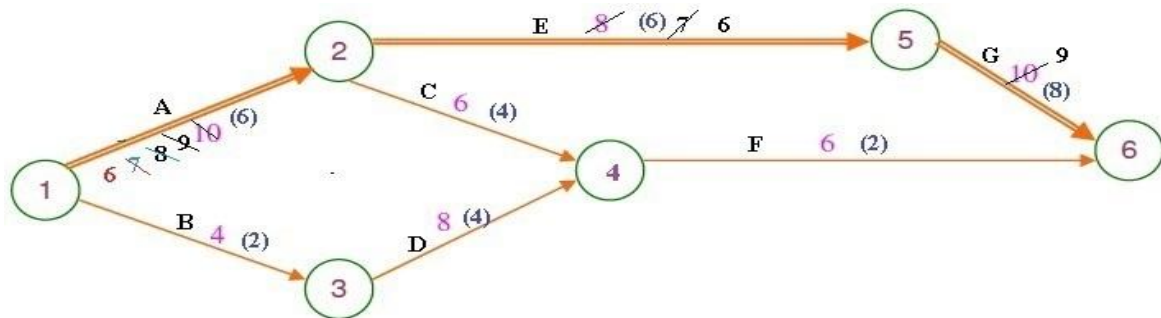


New critical path is 1-2-5-6 and Duration is  $6+6+10 = 22$  days

Activity A & E are crashed to their minimum duration; hence they cannot be crashed further.

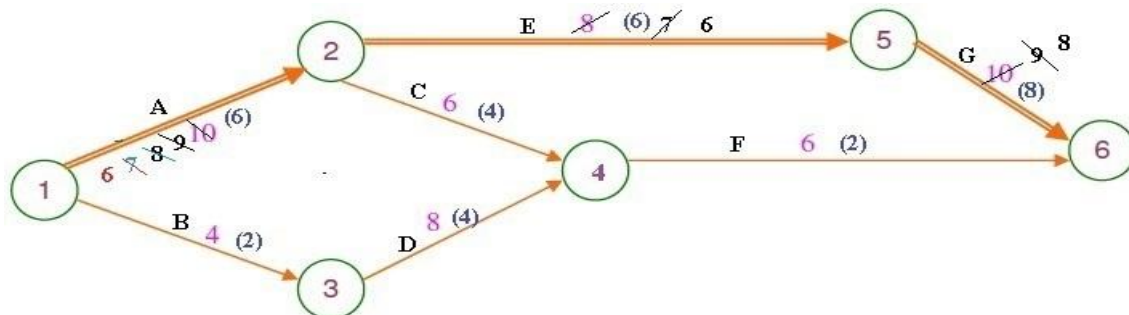
Now activity G only available for crashing

**Crashing-7:** Activity-G crashing by 1 day



New critical path is 1-2-5-6 and Duration is  $6+6+9 = 21$  days

**Crashing-8:** Activity-G crashing by 1 another day



New critical path is 1-2-5-6 and Duration is  $6+6+8 = 20$  days

The project has reached to its minimum duration of 20 days, so the project cannot be crashed further

**Cost Trade-off Matrix**

| Project duration | Direct Cost     |                      | Indirect Cost (C) | Total Cost (a + b + c) |
|------------------|-----------------|----------------------|-------------------|------------------------|
|                  | Normal cost (a) | Crash cost (b)       |                   |                        |
| 28 days          | 3700            | Nil                  | 2800              | 6500                   |
| 27 days          | 3700            | 50 = 50              | 2700              | 6450                   |
| 26 days          | 3700            | 50+50 = 100          | 2600              | 6400                   |
| 25 days          | 3700            | 100+50 = 150         | 2500              | 6350                   |
| <b>24 days</b>   | <b>3700</b>     | <b>150+50 = 200</b>  | <b>2400</b>       | <b>6300</b>            |
| <b>23 days</b>   | <b>3700</b>     | <b>200+100 = 300</b> | <b>2300</b>       | <b>6300</b>            |
| <b>22 days</b>   | <b>3700</b>     | <b>300+100 = 400</b> | <b>2200</b>       | <b>6300</b>            |
| 21 days          | 3700            | 400+130 = 530        | 2100              | 6330                   |
| 20 days          | 3700            | 530+130 = 660        | 2000              | 6360                   |

**Optimum project duration is 22 days and optimum project cost is 6300**

2. A project has 5 activities and the related information given below.

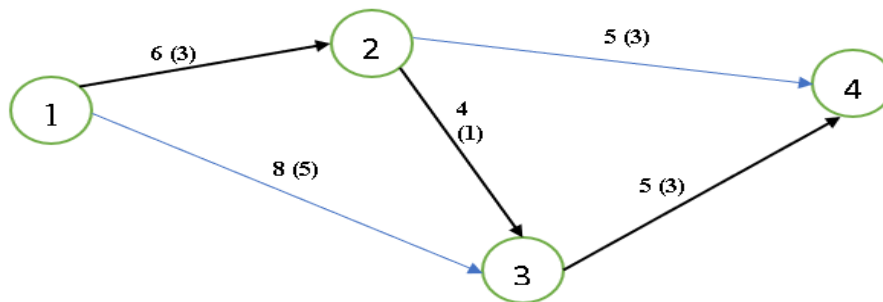
| Activity | Normal Time<br>(weeks) | Crash Time<br>(weeks) | Normal<br>Cost | Crash<br>Cost |
|----------|------------------------|-----------------------|----------------|---------------|
| 1-2      | 6                      | 3                     | 7,000          | 14,500        |
| 1-3      | 8                      | 5                     | 4,000          | 8,500         |
| 2-3      | 4                      | 1                     | 6,000          | 9,000         |
| 2-4      | 5                      | 3                     | 8,000          | 15,000        |
| 3-4      | 5                      | 3                     | 5,000          | 11,000        |

Indirect cost works out to be 3000 per week

Find out the optimum project cost and optimum project duration

**Solution:**

**Network Construction:**



*Paths of the network are*

1-2-4 = 6+5 = 11 weeks

1-2-3-4 = 6+4+5 = 15 weeks

1-3-4 = 8+5 = 13 weeks

**Critical path is 1-2-3-4 = 6+4+5 = 15 weeks**

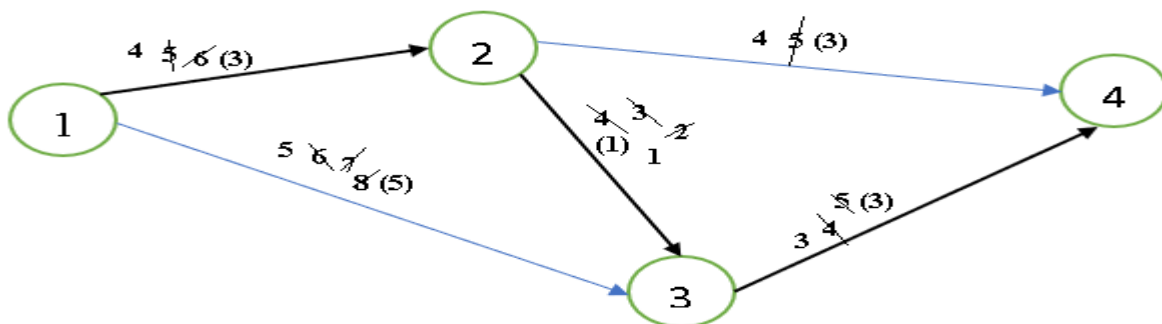
Project duration is 15 weeks and the **crash duration of the project is 3+1+3 = 7 weeks**

**Calculation of Cost Slope or Crash Cost:**

| Activity | Normal<br>Time | Crash<br>Time | Normal<br>Cost | Crash<br>Cost | Cost Slope = $\frac{C_c - N_c}{N_T - C_T}$ |
|----------|----------------|---------------|----------------|---------------|--------------------------------------------|
| 1-2      | 6              | 3             | 7,000          | 14,500        | 2,500                                      |
| 1-3      | 8              | 5             | 4,000          | 8,500         | 1,500                                      |
| 2-3      | 4              | 1             | 6,000          | 9,000         | 1,000                                      |
| 2-4      | 5              | 3             | 8,000          | 15,000        | 3,500                                      |
| 3-4      | 5              | 3             | 5,000          | 11,000        | 3,000                                      |

**Project Crashing table:**

| Crashing No.                    | Activities available for crashing | Cost slope                   | Decision   | Project duration | Critical Path              |
|---------------------------------|-----------------------------------|------------------------------|------------|------------------|----------------------------|
| 1                               | 1-2<br>2-3<br>3-4                 | 2500<br>1000<br>3000         | 2-3        | 14 weeks         | 1-2-3-4                    |
| 2                               | 1-2<br>2-3<br>3-4                 | 2500<br>1000<br>3000         | 2-3        | 13 weeks         | 1-2-3-4<br>1-3-4           |
| 3                               | 1-2<br>2-3<br>3-4<br>1-3          | 2500<br>1000<br>3000<br>1500 | 1-3<br>2-3 | 12 weeks         | 1-2-3-4<br>1-3-4           |
| 4                               | 1-2<br>1-3<br>3-4                 | 2500<br>1500<br>3000         | 3-4        | 11 weeks         | 1-2-3-4<br>1-3-4<br>1-2--4 |
| 5                               | 1-2<br>1-3<br>2-4<br>3-4          | 2500<br>1500<br>3500<br>3000 | 1-2<br>1-3 | 10 weeks         | 1-2-3-4<br>1-3-4<br>1-2--4 |
| 6                               | 1-2<br>1-3<br>2-4<br>3-4          | 2500<br>1500<br>3500<br>3000 | 1-2<br>1-3 | 09 weeks         | 1-2-3-4<br>1-3-4<br>1-2--4 |
| 7                               | 1-2<br>2-4<br>3-4                 | 2500<br>3500<br>3000         | 2-4<br>3-4 | 08 weeks         | 1-2-4<br>1-3-4             |
| No further crashing can be done |                                   |                              |            |                  |                            |



### Cost Trade-off Matrix

| Project duration | Direct Cost     |                | Indirect Cost (C) | Total Cost (a + b + c) |
|------------------|-----------------|----------------|-------------------|------------------------|
|                  | Normal cost (a) | Crash cost (b) |                   |                        |
| 15 weeks         | 30,000          | Nil            | 45,000            | 75,000                 |
| 14 weeks         | 30,000          | 1,000          | 42,000            | 73,000                 |
| 13 weeks         | 30,000          | 2,000          | 39,000            | 71,000                 |
| 12 weeks         | 30,000          | 4,500          | 36,000            | 70,500                 |
| <b>11 weeks</b>  | <b>30,000</b>   | <b>7,500</b>   | <b>33,000</b>     | <b>70,500</b>          |
| 10 weeks         | 30,000          | 11,500         | 30,000            | 71,500                 |
| 09 weeks         | 30,000          | 15,500         | 27,000            | 72,500                 |
| 08 weeks         | 30,000          | 21,000         | 24,000            | 75,000                 |

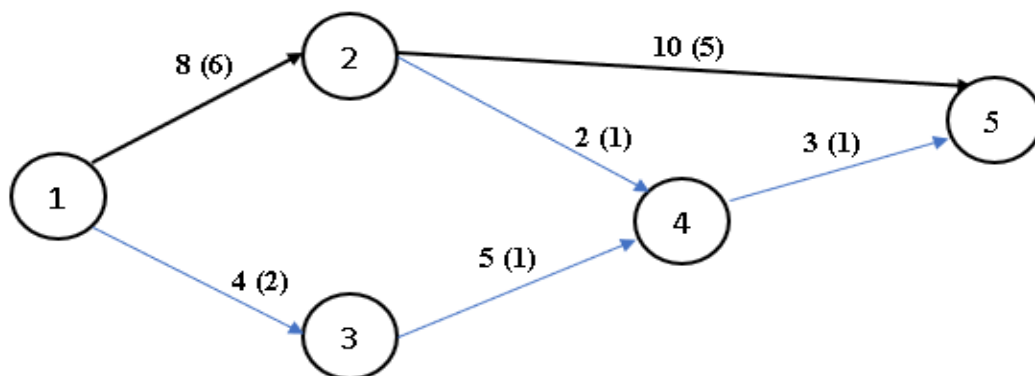
**Optimum project duration is 11 weeks and optimum project cost is 70,500**

3. From the activity details given below, determine the optimal project duration and optimal project cost. Indirect cost (slope) is Rs. 70 per day

| Activity | Normal      |      | Crash       |      |
|----------|-------------|------|-------------|------|
|          | Time (days) | Cost | Time (days) | Cost |
| 1-2      | 8           | 100  | 6           | 200  |
| 1-3      | 4           | 150  | 2           | 350  |
| 2-4      | 2           | 50   | 1           | 90   |
| 2-5      | 10          | 100  | 5           | 400  |
| 3-4      | 5           | 100  | 1           | 200  |
| 4-5      | 3           | 80   | 1           | 100  |

**Answer:**

### Network Construction



*Paths of the network are:*

$$1-2-5 = 8+10 = 18 \text{ days}$$

$$1-2-4-5 = 8+2+3 = 13 \text{ days}$$

$$1-3-4-5 = 4+5+3 = 12 \text{ days}$$

Critical path is **1-2-5** and Project duration is **18 days**

Crash (minimum) duration of the project is  $6+5 = 11 \text{ days}$

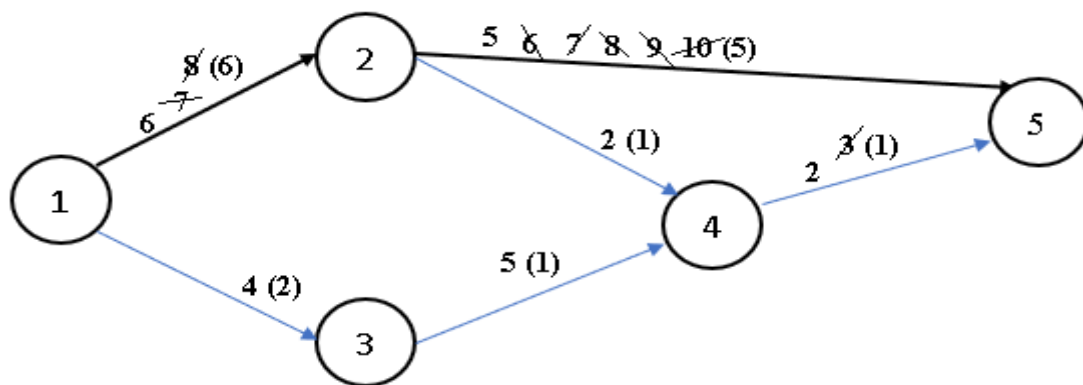
### Calculation of Cost Slope or Crash Cost:

| Activity | Normal Time | Crash Time | Normal Cost | Crash Cost | Cost Slope = $\frac{C_c - N_c}{N_T - C_T}$ |
|----------|-------------|------------|-------------|------------|--------------------------------------------|
| 1-2      | 8           | 6          | 100         | 200        | 50                                         |
| 1-3      | 4           | 2          | 150         | 350        | 100                                        |
| 2-4      | 2           | 1          | 50          | 90         | 40                                         |
| 2-5      | 10          | 5          | 100         | 400        | 60                                         |
| 3-4      | 5           | 1          | 100         | 200        | 25                                         |
| 4-5      | 3           | 1          | 80          | 100        | 10                                         |

### Project Crashing Table:

| Crashing No. | Activities available for crashing | Cost slope | Decision | Project duration | Critical Path    |
|--------------|-----------------------------------|------------|----------|------------------|------------------|
| 1            | 1-2                               | 50         | 1-2      | 17 days          | 1-2-5            |
|              | 2-5                               | 60         |          |                  |                  |
| 2            | 1-2                               | 50         | 1-2      | 16 days          | 1-2-5            |
|              | 2-5                               | 60         |          |                  |                  |
| 3            | 2-5                               | 60         | 2-5      | 15 days          | 1-2-5            |
| 4            | 2-5                               | 60         | 2-5      | 14 days          | 1-2-5            |
| 5            | 2-5                               | 60         | 2-5      | 13 days          | 1-2-5            |
| 6            | 2-5                               | 60         | 2-5      | 12 days          | 1-2-5<br>1-3-4-5 |
| 7            | 2-5                               | 60         | 2-5      | 11 days          | 1-2-5<br>1-3-4-5 |
|              | 1-3                               | 100        |          |                  |                  |
|              | 3-4                               | 25         |          |                  |                  |
|              | 4-5                               | 10         |          |                  |                  |

No further crashing can be done as project duration reached to its minimum of 11 days



### Exercise Problems

1. Consider the following project with the estimates of time in weeks:

| Job | Preceding Activities | $t_o$ | $t_m$ | $t_p$ |
|-----|----------------------|-------|-------|-------|
| A   | -                    | 2     | 4     | 6     |
| B   | -                    | 3     | 5     | 7     |
| C   | A                    | 5     | 6     | 13    |
| D   | A                    | 4     | 8     | 12    |
| E   | B, C                 | 5     | 6     | 13    |
| F   | D, E                 | 6     | 8     | 14    |

Find the probability that the project will be completed in 27 weeks.

2. The following table shows the activities of a construction project and other data.

| Activity | Normal      |      | Crash       |      |
|----------|-------------|------|-------------|------|
|          | Time (days) | Cost | Time (days) | Cost |
| 1-2      | 6           | 50   | 4           | 80   |
| 1-3      | 5           | 80   | 3           | 150  |
| 2-4      | 5           | 60   | 2           | 90   |
| 2-5      | 8           | 100  | 6           | 300  |
| 3-4      | 5           | 140  | 2           | 200  |
| 4-5      | 2           | 60   | 1           | 80   |

Indirect cost is Rs 20 per day. Crash the activities to find the optimal duration of the project and optimal project cost.

### **Differences between PERT and CPM**

| CPM                                                                                                                         | PERT                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| 1. It is activity oriented                                                                                                  | 1. It is event oriented                                                                         |
| 2. It is a deterministic model with one time estimate                                                                       | 2. It is a probabilistic model with three-time estimates                                        |
| 3. Its terminology like arrow diagram, nodes and floats                                                                     | 3. Its terminology like network diagram, events and slack                                       |
| 4. In CPM cost is not directly proportional to time                                                                         | 4. In PERT cost varies directly with the time                                                   |
| 5. In CPM cost is the controlling factor                                                                                    | 5. In PERT time is the controlling factor                                                       |
| 6. In CPM critical path passes through critical activities                                                                  | 6. In PERT critical path passes through critical events                                         |
| 7. CPM is employed to those projects where overall cost is of primary importance                                            | 7. PERT used in project where resources are available as and when required                      |
| 8. It is suitable for problems in industrial setting, plant maintenance, civil construction, project expansion schemes etc. | 8. It is suitable in defense projects and R&D where activity times cannot be reliably predicted |